

Sanya Kapoor

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Profile Summary

MSc in Data Science with experience as a Cloud & Networking Engineer. Skilled in Python, SQL, Power BI, cloud infrastructure, and large-scale data systems. Experienced in analytics, ETL pipelines, KPI reporting, and automation, with growing expertise in machine learning and predictive analytics.

Technical Skills

Programming & Analytics: Python (Pandas, NumPy, scikit-learn), SQL, R

Visualization & BI: Power BI, Tableau, Excel, R Markdown

Data Engineering: ETL Pipelines, Data Integration, Monitoring, Metric Extraction

Cloud & Automation: AWS, Terraform Cloud, n8n

Techniques: A/B Testing, Statistical Analysis, Forecasting, Feature Engineering, Exploratory Data Analysis, Predictive Analytics

Work Experience

Tata Communications, Chennai, India

Jul 2023 – Mar 2025

Global IP and Ethernet Engineer

- Built automated monitoring systems using Python and REST APIs across 50+ network devices, reducing reporting time by 30%.
- Designed Power BI dashboards with 12 KPIs using SQL Direct-query and Oracle databases, improving analytics adoption by 25%.
- Analysed 500 GB+ traffic datasets using Python to identify bottlenecks and optimise peak traffic performance.
- Developed routing optimisation models that improved network reliability by 20%.
- Built a conversational Infrastructure-as-Code pipeline using Terraform Cloud and n8n for AWS provisioning automation.

Mentor Minds

May 2023

Data Analyst Intern

- Generated weekly performance reports in Power BI, tracking 8 key business metrics.
- Designed Python ETL pipelines integrating customer behaviour data from 3 sources, reducing preparation time by 60%.
- Conducted quantitative and qualitative analysis to identify user behaviour trends contributing to business improvements.
- Collaborated with stakeholders to define KPIs and build interactive dashboards with drill-down functionality.

Education

University of Bristol, United Kingdom

Sept 2025 - Sept 2026

MSc Data Science

Coursework: Large-Scale Data Engineering, AI and Text Analytics, Visual Analytics, Statistical Computing and Empirical Modelling, Machine Learning

SRM Institute of Science & Technology, Chennai, India

June 2019 - May 2023

Bachelor of Technology, Electronics & Communication Engineering

GPA: 9.21/10

Projects & Research

Detection of Renal Disease and Health Optimization Tracker

- Developed machine learning models, including SVM, KNN, and Naive Bayes for kidney disease prediction.
- Built CNN-based CT scan segmentation models and integrated a Firebase-connected Android health tracking app.
- Research published in IEEE.

Extracting Structured Clinical Trial From Biomedical Structured Clinical Trial Information From Biomedical Abstracts Using EBM-NLP

- Compared nine extraction systems along two design axes namely rule-based versus LLM-based, and end-to-end versus decomposed, on a locked 50-document evaluation set, and report token overlap F1, strict span F1, coverage and downstream keyword query usability.

Predicting Customer Retention Using Machine Learning

- Predicted customers at risk of churning and enable proactive retention strategies.
- Leverages classification models, feature engineering, and model evaluation techniques, and is deployed using an interactive web application